

ALLinHELP — Developer Discovery Notes

From: Backend Developer
To: ALLinHELP Product Team
Date: September 24, 2026
Re: PRD v1.0 + SAD v1.0 Review — Pre-Development Alignment

1. Introduction

Thank you for sharing the PRD and System Architecture Document. Both documents are well-structured and reflect a clear, ambitious product vision. This note is not a rejection of that vision — it is a developer's honest assessment to ensure we deliver something *real and valuable* within the agreed 3-month window, rather than an incomplete version of something larger.

Everything documented below is intended to protect the project's success. I am raising these points **before writing a single line of code**, because resolving them now is 10x cheaper than resolving them at month 2.

2. Development Team & Capacity

Role	Capacity
UI/UX Designer	Full-time
Frontend Developer	Full-time
Backend Developer	Full time

This is a workable team for a well-scoped MVP. It is **not sufficient** for the full PRD scope as written.

3. Scope Reality Check

The PRD MVP includes the following deliverables:

- 2 separate mobile applications (Customer App + Helper App)
- 1 web application
- 1 Admin Dashboard
- 11+ backend microservices
- 6 distinct AI engines (Matching, Pricing, Trust, Dispatch, Forecast, Fraud Detection)
- Real-time GPS tracking
- In-app messaging
- Digital wallet + payment processing
- Government ID + selfie identity verification

This scope represents an estimated 12–18 months of work for a team of this size.

Delivering all of the above in 3 months would result in either: (a) A partially functional system that is not production-ready, or
(b) A technically complete system with inadequate testing, documentation, or reliability

Neither outcome serves the client or the business.

4. Our Recommendations

4.1 — Single App Instead of Two Separate Apps

Proposal: Build **one application** where the displayed interface depends on whether the user registered as a **Customer** or a **Helper (Service Provider)**.

Reasoning:

- Both user types share the same core functionality: registration, authentication, profiles, messaging, payments, ratings
- The differentiation is in the *view and workflow*, not the underlying system
- This halves the frontend development effort
- This is a proven pattern used by platforms like Fiverr, TaskRabbit, and numerous others
- A single codebase is significantly easier to maintain and iterate on

Client question: Are there specific business or brand reasons requiring two completely separate applications?

4.2 — Start with a PWA Instead of Native Apps

Proposal: For the 3-month MVP, build a **Progressive Web App (PWA)** instead of native mobile apps (iOS/Android).

What is a PWA?

A PWA is a website that behaves like a mobile app. It installs to the home screen, works offline (partially), sends push notifications, and accesses device features like the camera and location. Examples of successful platforms using this model: **BetKing**, **Twitter Lite**, **Starbucks**, and many others.

Why PWA for ALLinHELP MVP?

Factor	PWA	Native App
Development time	✓ 1 codebase, faster	✗ Separate builds for iOS + Android
App Store approval	✓ None needed	✗ Apple review can take 1–4 weeks
Update deployment	✓ Instant	✗ Requires re-submission + user update
GPS/Location	✓ Supported	✓ Supported
Push Notifications	✓ Supported (iOS 16.4+)	✓ Supported
Camera (for ID)	✓ Supported	✓ Supported

Factor	PWA	Native App
Offline capability	△ Partial	✓ Full
Real-time tracking	△ Limited (polling)	✓ Background GPS
User trust/perception	△ Slightly lower	✓ Higher

PWA is the recommended MVP strategy because it allows us to deliver a working, testable product to real users without App Store delays or platform-specific development costs. Native apps can be built in Phase 2 based on validated user feedback from the PWA.

Client question: Is there a specific business requirement that mandates App Store presence at launch?

4.3 — Modular Monolith Instead of Full Microservices at Launch

The SAD proposes a full microservices architecture from day one. This is the *correct long-term architecture*, but it introduces significant overhead for a 3-month build:

- API gateway configuration
- Inter-service communication and failure handling
- Distributed logging and tracing
- Independent deployment pipelines per service
- Network latency between services

Recommendation: Build a **modular monolith** — code is organized into the exact service modules defined in the SAD (Auth, User, Helper, Booking, Matching, Pricing, etc.) but deployed as a single unit. This gives us:

- All the structural boundaries of microservices
- None of the operational overhead at launch
- A clear, low-risk path to extracting true microservices in Phase 2

This is the approach used by companies like Shopify, Stack Overflow, and Basecamp at early scale.

Client question: Is there a technical requirement that demands true microservices from day 1 (e.g., separate teams owning separate services)?

4.4 — AI Engines: Proposed Phase Allocation

The PRD lists 6 AI engines. We recommend the following phasing:

AI Engine	PRD Classification	Our Recommendation
AI Adaptive Matching Engine	MVP	✓ Phase 1 (simplified rule-based + basic ML)
AI Fair Price Prediction Engine	MVP	✓ Phase 1 (rule-based with configurable factors)
AI Trust Intelligence Engine	MVP	△ Phase 2 (use static rating average for MVP)

AI Engine	PRD Classification	Our Recommendation
AI Dispatch Engine	MVP	⚠ Phase 2 (manual ordering sufficient for MVP)
AI Demand Forecasting Engine	Deferred	✅ Phase 3 (as planned)
AI Fraud Detection Engine	MVP	⚠ Phase 2 (basic rules for MVP; full ML model later)

Rationale: A rule-based Matching engine (location + rating + availability + skills) + a rule-based Pricing engine (service type + location zone + time + demand factor) are sufficient to validate the marketplace model. Complex ML is high-effort and requires real transaction data to train anyway — which we won't have at launch.

5. Proposed MVP Scope (Revised)

✅ Included in 3-Month MVP

- ☐ Single PWA (Customer + Helper views, role-based)
- ☐ User registration, login, and authentication (JWT + OTP)
- ☐ Identity verification (Government ID upload + admin review queue)
- ☐ Service category management (admin-configurable)
- ☐ Booking request creation and lifecycle management
- ☐ GPS-based helper discovery (location search radius)
- ☐ AI Matching Engine v1 (rule-based: proximity + rating + skills + availability)
- ☐ AI Pricing Engine v1 (rule-based: service type + location + time + urgency)
- ☐ In-app chat (real-time messaging)
- ☐ Payments (Paystack integration)
- ☐ Digital wallet (balance, transactions, withdrawal requests)
- ☐ Ratings and reviews (post-booking)
- ☐ Push notifications (booking updates, messages)
- ☐ Admin dashboard (user management, helper verification, service config, basic analytics)

❌ Deferred to Phase 2 (Month 4–6)

- Native iOS + Android apps
- Real-time background GPS tracking
- AI Trust Intelligence Engine
- AI Fraud Detection Engine (full ML)
- AI Dispatch Engine (optimized)
- Voice messaging
- Advanced analytics and BI dashboards

❌ Deferred to Phase 3 (Month 7+)

- AI Demand Forecasting
- Enterprise APIs

- Multi-language AI assistant
- Smart contracts / Insurance integration
- Predictive scheduling

6. Questions Requiring Client Response

Please respond to the following before development begins. These decisions directly affect architecture and delivery timelines.

1. **Platform:** Are you open to launching as a PWA for MVP, with native apps in Phase 2?
2. **App structure:** Can the Customer and Helper experiences live in one app (role-based views) rather than two separate apps?
3. **AI scope:** Do you accept the proposed phasing of AI engines (Matching + Pricing in Phase 1 only)?
4. **Identity verification:** Do you have a preferred third-party verification provider, or should we recommend one (e.g., Smile Identity, Dojah)?
5. **Payments:** Confirm the target launch market — Paystack for Nigeria, Flutterwave for broader Africa?
6. **Definition of "done":** Is the 3-month deliverable expected to be a production launch, or a beta/soft launch for testing?
7. **Design assets:** Is the UI/UX designer working from an existing design system or starting from scratch?
8. **Real-time GPS:** Is live GPS tracking of helpers during a job a hard MVP requirement, or can status updates (En Route → Arrived → In Progress) suffice for launch?
9. **Data and legal:** Is a Privacy Policy and Terms of Service already drafted? (Required for any payment or identity processing)

7. Proposed Development Approach

Element	Decision	Rationale
Backend framework	FastAPI (Python)	Async, AI/ML-native ecosystem, auto-generated docs
Database	PostgreSQL	As specified in SAD — correct choice
Cache	Redis	As specified in SAD
Frontend	React (PWA) or React Native Web	Single codebase across platforms
Payments	Paystack	Best DX for African market
File storage	AWS S3 or Cloudflare R2	Profile images, ID documents
Push notifications	Firebase Cloud Messaging (FCM)	Free, cross-platform
Real-time chat	Socket.io	Simple, reliable, well-documented
Deployment architecture	Modular monolith → microservices (Phase 2)	Right for timeline and team size

8. Proposed Timeline (Pending Your Responses)

Milestone	Target	Deliverable
M0 — Alignment	Week 1	Agreed scope, tech stack, design kickoff
M1 — Foundation	End of Month 1	Auth, Users, Helpers, Bookings, Admin Panel
M2 — Core Flow	End of Month 2	Full booking flow, AI matching, payments, chat
M3 — Launch Ready	End of Month 3	PWA deployed, tested, identity verification, ratings

9. Next Steps

- 1. Client reviews this document and responds to questions in Section 6
- 2. Alignment call scheduled (recommended: within 5 business days)
- 3. Final MVP scope document produced and signed off by both parties
- 4. Week 1 development kickoff begins

All scope changes after the aligned MVP document is signed will be evaluated as change requests and may affect the delivery timeline.

This document was prepared in good faith to ensure project success. We are committed to delivering a high-quality, production-ready MVP within the agreed timeline, provided scope is aligned as proposed.

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